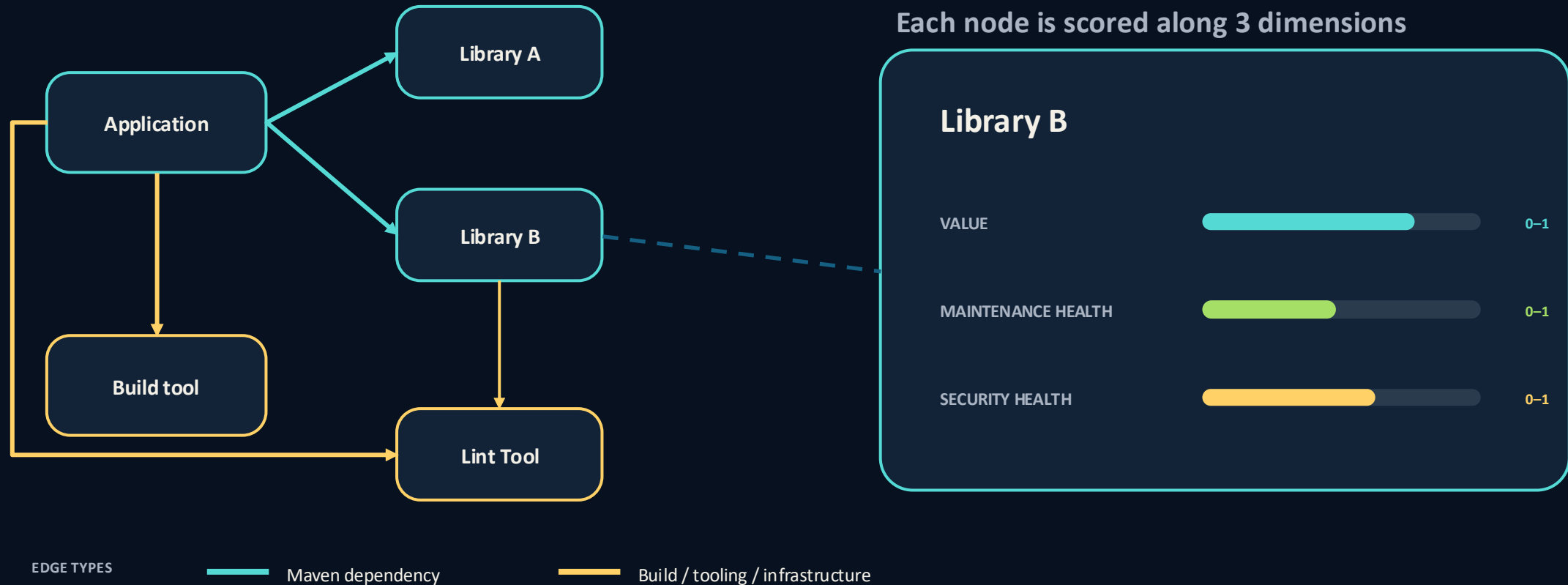


Scala Ecosystem Security Model

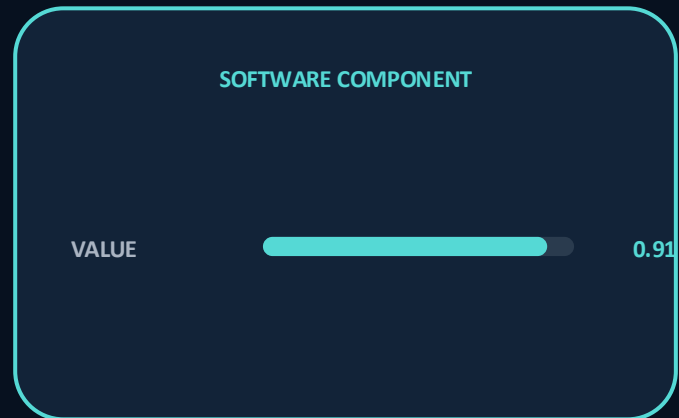
Ecosystem is a Graph



Nodes are software components; edges are software dependencies.

Risk Pattern

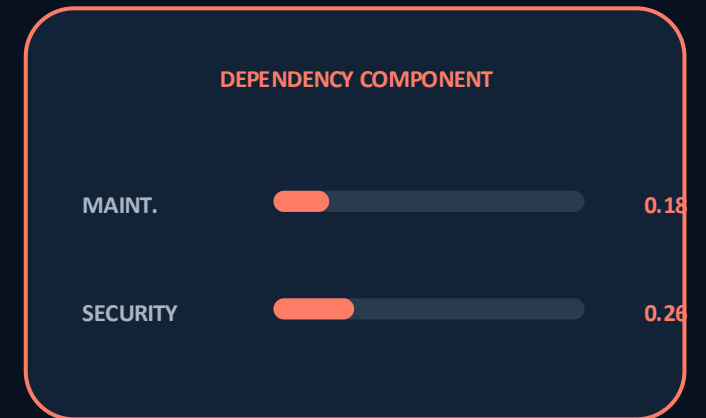
HIGH VALUE



DEPENDS ON

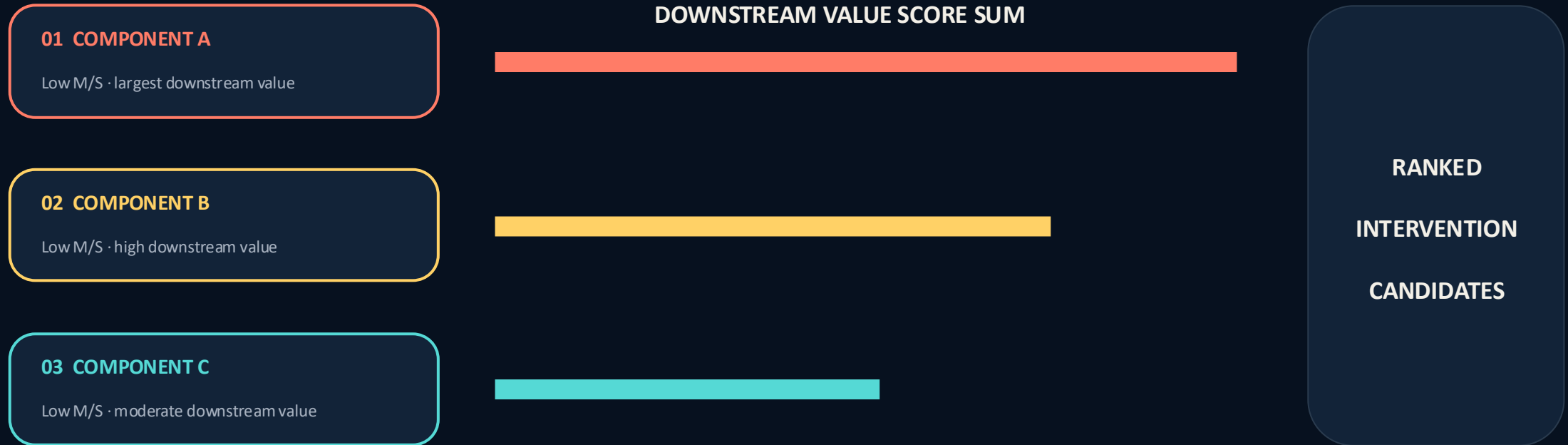


LOW MAINTENANCE OR SECURITY



Model Output

Weak components (low M/S) ranked by downstream value exposed (sum of dependants' V)



Downstream value is calculated from the dependency graph.

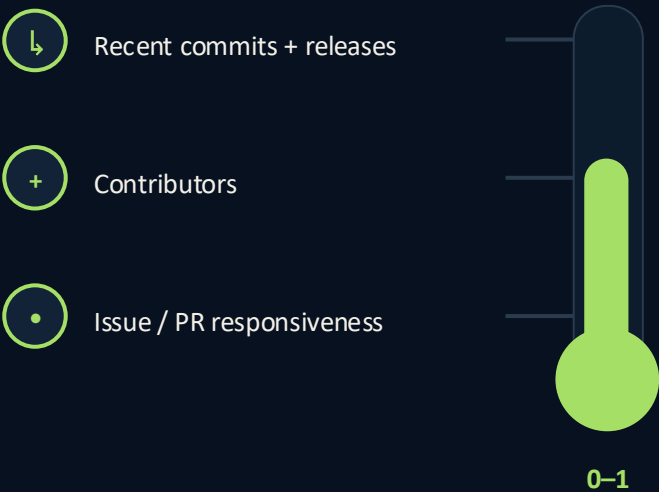
Scoring Model

Each dimension uses an adapted [OpenSSF Criticality Score](#) mechanism with its own inputs, weights and thresholds.

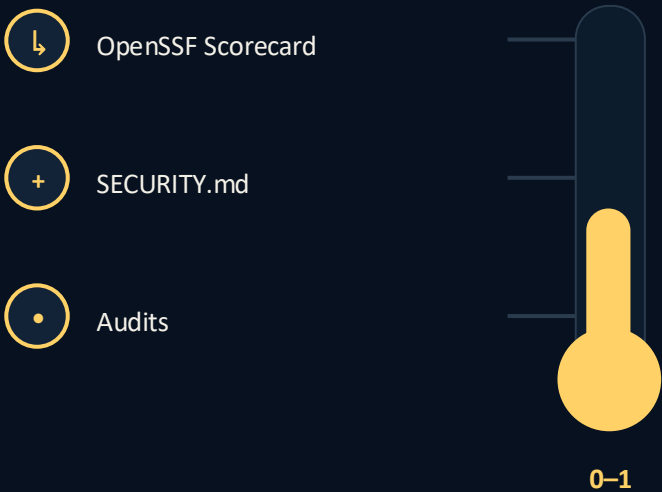
VALUE



MAINTENANCE HEALTH



SECURITY HEALTH



SCORING FORMULA [OpenSSF Criticality Score](#), with custom weights and inputs relevant to the particular dimension

SECURITY INPUT [OpenSSF Scorecard](#)

HUMAN-SUPPLIED INPUTS Security audit records, industry usage evidence

Exact inputs, weights and thresholds are provisional.

Iterative Model Design Workflow

4–6 WEEKS

